

Sruthi Mani

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PROFESSIONAL SUMMARY

Human Factors Engineering graduate student with experience supporting complex, safety-critical systems across healthcare, consumer products, and industrial environments. Skilled in planning and executing usability studies, identifying and mitigating use-related risks, and translating user needs into design requirements. Experienced in human-centered design across the full product lifecycle, from early concept through validation.

SKILLS

Human Factors & Usability: Usability Testing, Task Analysis (HTA, CTA, MTA), Hazard & Risk Analysis, URRR, RRA, Known Problem Analysis, User Needs Analysis, Accessibility & Inclusive Design, Cognitive Workload Assessment, MAUDE Database

Tools & Technical: Figma, SQL, Postman, APIs, Python, R Studio, Java, SolidWorks CAD, Onshape, Tableau, SAP, JIRA, Google Cloud Systems, Prometheus, Kibana, HTML

Languages: Tamil

EXPERIENCE

Human Factors Intern

May 2026 – Present

MITRE Corporation | Bedford, MA

DoD Sponsored Project

- Planned, executed, and synthesized an in-person usability study for a military hardware device, moderating sessions with 8 users and capturing behavioral and qualitative data from user interaction and navigation.
- Authored a formal usability report synthesizing user feedback into prioritized findings and design recommendations for the device.
- Led a card sort UX activity to categorize GPS functionality across basic vs. advanced user profiles, informing information architecture and feature tiering decisions.

FCC Sponsored Project

- Co-authored a 10-page summary brief synthesizing research from Deaf, Hard of Hearing, and DeafBlind communities on barriers to 911 access; report was formally submitted to the FCC with recommendations for improving emergency communication services.
- Coordinated participant recruitment for a usability study evaluating automatic phone captioning services, sourcing users to provide voice data for accessibility testing of emergency communication tools.

Research Assistant – Applied Cognition Lab

January 2026 – Present

Tufts University | Medford, MA

The lab investigates the interaction between technology and cognition with a focus on multitasking in real and simulated environments.

- Leading development of an HTML web-based stroke toolkit using Claude to help survivors navigate their recovery process in a streamlined way.
- Write and conduct user interviews with technical experts and stroke patients to ensure UI heuristics are applied to the toolkit.
- Arranging equipment and protocols and supervising participant engagement to enable accurate data collection.

Researcher – IDEA Lab Dentistry Project

January 2026 – Present

Tufts University | Medford, MA

- Working with a team to support designing an ergonomic dental grip to minimize pain associated with micro-movements that occur with dental tools, with a focus on the explorer tool.
- Using anthropometric data in 35th/65th percentile of female hands to form design decisions of the dental tools.

- Conducting background interviews, task analyses, user interviews, competitive analysis, and usability testing in collaboration with Tufts Dental School.

Associate Product Consultant

August 2023 – June 2025

Manhattan Associates | Atlanta, GA

- Led human-centered design activities for Cardinal Health warehouse management systems used in operational environments.
- Conducted formative usability testing, task walkthroughs, and worker surveys to redesign RF device screens, improving scanning accuracy, reducing task load, and increasing user efficiency and picks per minute.
- Performed task analyses, user interviews, and contextual inquiries to identify critical steps, user errors, and potential usability risks across warehouse fulfillment workflows.
- Collaborated with cross-functional teams, including internal leadership and R&D, using JIRA to prioritize and manage development and deployment of new features.
- Developed interactive prototypes and detailed mod specs capturing use-related hazards, supporting engineering teams in redesigning higher-risk workflows.
- Integrated WMS with Material Handling Equipment (PSI), improving label accuracy and reducing label failure rate by 20%, demonstrating effective human-system integration analysis.
- Recognized as one of the Top 20 Associates out of 300 in 2024 for outstanding contributions to client satisfaction and cross-functional collaboration.

Global Leadership Development Internship

May 2022 – August 2022

Whirlpool | Benton Harbor, MI

- Consolidated supplier and SAP data into a unified database to support engineering decision-making and reduce variability in component sourcing.
- Designed Tableau dashboards presenting cost-saving opportunities using HF principles for cognitive clarity.
- Awarded 1st Place – Whirlpool Product Development Innovation for designing an ADA-compliant dishwasher rack insert, demonstrating strong capabilities in human factors for accessibility and inclusive design.

Engineering and Lab Intern

June 2021 – August 2021

Consolidated Precision Products | Syracuse, NY

- Created and implemented a new workflow for porosity alloy testing procedures in the aerospace manufacturing industry.
- Developed an innovative solution to remove contaminants from ceramic molds for castings, preserving and enhancing the quality of metal parts produced.
- Performed daily pH, temperature, and binder solids tests, ensuring stable material conditions and supporting risk-based process control.

PROJECTS

SharkNinja Innovation Challenge – 2nd Place

October 2025 – Present

SharkNinja | Boston, MA

- Selected as a finalist in SharkNinja's national innovation challenge; developed Matcha Luxe, a consumer wellness appliance concept grounded in human-centered design. Awarded 2nd place and \$15,000 in product development funding.
- Conducted user needs analysis, market research, and interviews with mentorship from SharkNinja product and innovation leaders.
- Delivered a live product pitch to SharkNinja executives and industry experts, with exposure to early-stage patent and product commercialization considerations.

Senior Research Project – Ergonomic Analysis

January 2023 – May 2023

Andritz Association | Muncy, PA

- Conducted a comprehensive NIOSH lifting and biomechanical analysis to identify high-strain tasks and evaluate risk-related use scenarios.
- Identified lifting-height hazards contributing to back injuries and proposed redesign of manufacturing workflow.
- Designed a loading ramp intervention that reduced injury likelihood and improved task safety, ergonomic load, and training requirements.

EDUCATION

M.S. Human Factors Engineering

Present – December 2026

Tufts University | Medford, MA

- Relevant Coursework: Human-Machine Systems, Product Design & Usability, Medical Technology Development

B.S. Industrial Engineering

September 2019 – May 2023

The Pennsylvania State University | University Park, PA

- Awards: NYSLI South Korea Study Abroad Scholarship 2019; People's Choice Award, Engineering Design Competition 2020; Nelson/McGraw Foundation Industrial Engineering Scholarship 2022